

# Harshal Abdulla

Associate / graduate software engineer · backend-leaning full stack

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MSc Computer Science graduate who builds systems where being wrong is expensive. I built and maintain a commercial ordering system that takes real money in a restaurant every day, and wrote the property-based test suite that proves its pricing is correct. Looking for an associate or graduate software engineer role in Ireland. Available now.

## SELECTED PROJECTS

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### Restaurant ordering and till system

Commercial, sole developer · 2026

Client system running daily in a restaurant in Co. Kildare

- Built four surfaces (customer kiosk, kitchen till, table service, admin) from one Expo codebase against one Postgres database, in seven weeks, as the only developer.
- Put pricing and entitlement rules in plpgsql behind *security definer* so all four surfaces get the same answer and no client can bypass the till. Around 70 Postgres functions behind row level security across 94 migrations.
- Removed the application server entirely: static hosting plus row level security, with the client talking to Postgres through PostgREST and RLS as the security boundary.
- Money is integer euro cents with no floating point in pricing. Order lines are immutable snapshots and corrections are soft deletes, so history is never rewritten.
- Wrote a property-based test suite from scratch with no test framework and no dependencies: 43 properties, 560,015 generated cases per run in about a second, seeded so any failure replays exactly.
- Measured an order-dependent pricing effect across 20,000 baskets, showed the owner the numbers, and locked the guarantee it rests on into a test that runs on every change.
- 154 commits. 15,109 lines of TypeScript. 251 orders and 1,474 dishes in August 2026.

**Stack:** TypeScript 5.9, React 19.2, React Native, Expo SDK 54, Zustand, PostgreSQL via Supabase, row level security, pg\_cron ·  
Demo: [order-dependent-pricing.surge.sh](https://order-dependent-pricing.surge.sh)

### Fault-tolerant notification system

Personal · in progress, around 80%

- Distributed delivery pipeline built so a message is never silently lost and never sent twice, even when the broker, database or consumer crashes mid-flight.
- Producer writes a PENDING row to Postgres *before* publishing to Kafka, so a crash between the two leaves a durable record that a recovery sweep picks up.
- Consumer reads row status before acting (SELECT before UPDATE), which makes it idempotent by design and absorbs Kafka's at-least-once duplicates as a no-op.
- Exponential backoff with a dead-letter queue, and retry counters held in Redis rather than process memory so restarts and parallel consumers cannot break the count.

**Stack:** Python, Apache Kafka, PostgreSQL, Redis, Docker Compose · [github.com/Harshal-Abdulla/Kafka-Notification-System](https://github.com/Harshal-Abdulla/Kafka-Notification-System)

### Sketchpad Retro

Individual MSc project · 2025

- Browser rebuild of Ivan Sutherland's 1963 Sketchpad, with the constraint solver, geometry engine and undo/redo stack all written from scratch and the interface drawn in SVG.
- Constraints including parallelism, perpendicularity, equal length and fixed distance, plus snapping, zooming, symbols and instances, and state persisted via JSON and local storage.

**Stack:** Next.js 14, React, TypeScript, SVG

### Twix

Group MSc project, team of five · 2025

- Led a team of five building a real-time chat application. My own share of the work was the interface and the testing.
- Based the layout on Twitter so all five of us had a single reference to build against, rather than designing the interface from scratch and arguing about it.
- Schema designed to stay consistent with several people connected and sending at once.

**Stack:** Java, WebSockets, PostgreSQL, React, TypeScript, Vite, Axios, Bootstrap

SyncUp

Group MSc project · 2025

- Kanban task tool: register, log in, and move tasks across To Do, In Progress and Done. Full CRUD over a REST API, with password hashing using bcryptjs.

Stack: Node.js, Express, MongoDB, React

EXPERIENCE

Frontend Engineer, Cofount Labs LLP

September 2023 – August 2024 · Remote

- Built and shipped client-facing web applications for agency clients.

EDUCATION

MSc Computer Science (Software Engineering)

2024 – 2025

Maynooth University, Ireland · First Class Honours

Software testing and quality assurance, requirements engineering and system design, advanced object-oriented programming and scalable systems, distributed systems, applied cryptography, deep learning foundations.

B.Tech Computer Science and Technology

2019 – 2023

Jyothi Engineering College, Kerala, India

CERTIFICATIONS

Microsoft Certified: Azure Fundamentals (AZ-900)

March 2026

TECHNICAL SKILLS

|                |                                                                        |
|----------------|------------------------------------------------------------------------|
| Languages      | TypeScript, JavaScript, Python, SQL                                    |
| Frontend       | React, React Native, Next.js, Expo, Zustand, Tailwind CSS              |
| Backend & data | Node.js, Express, PostgreSQL, Supabase, MongoDB, Redis, REST           |
| Infrastructure | Apache Kafka, Docker Compose, GitHub Actions, Git                      |
| Testing        | Property-based testing, test design, row level security policy testing |